

Lab 3: Multi-Source Retail Sales Data Integration and Analysis

Duration: 2 Hours

Platform: R / Google Colab

Domain: Retail Analytics

Problem Statement

A multinational retail company maintains its business data across multiple systems. Sales transactions are stored in CSV format, product information in JSON format, and customer information in Excel format. Management wants to integrate these datasets to obtain a consolidated view of sales performance and customer value.

As a Data Analyst, develop an R-based solution to import, clean, integrate, and analyze these heterogeneous datasets. The final processed data should also be stored in an SQL database for future analysis.

Dataset

Use the UCI Online Retail Dataset containing transaction-level retail sales information.

The supplied data should be organized into:

- transactions.csv – InvoiceNo, StockCode, CustomerID, Quantity, InvoiceDate
- products.json – StockCode, Description, UnitPrice
- customers.xlsx – CustomerID, Country

Task 1: Import and Clean the Data

Dataset: [UCI Machine Learning Repository – Online Retail Dataset](#)

Import the CSV, JSON, and Excel datasets into R using suitable packages.

Inspect the datasets and perform necessary cleaning for:

- Missing values
- Duplicate records
- Invalid or zero quantities
- Invalid or zero unit prices

Create a new attribute:

Revenue = Quantity × UnitPrice

Briefly mention the cleaning decisions taken.

Task 2: Integrate the Multiple Data Sources

Combine the transaction, product, and customer datasets using appropriate dplyr join operations such as `left_join()` or `inner_join()`.

After integration:

- Verify the dimensions of the final dataset.
- Identify unmatched records, if any.
- Justify the type of join selected.

Task 3: Perform Sales and Customer Analysis

Using appropriate dplyr operations, determine:

1. Total sales revenue.
2. Top 5 products based on revenue.
3. Top 5 countries based on revenue.
4. Top 5 customers based on total purchase value.

Using `case_when()`, classify customers into:

- Low Value
- Medium Value
- High Value
- Premium

Use suitable thresholds based on customer purchase values.

Finally, identify one high-performing market and one underperforming market and briefly justify your observations.

Task 4: Store and Retrieve Data Using SQL

Create a SQLite database and export the final cleaned and integrated dataset to a table named `retail_sales`.

Execute any two meaningful SQL queries from R, such as:

- Top 5 customers based on revenue.
- Total revenue by country.

Conclude the analysis with three important business insights obtained from the results.

Expected Deliverables

- 1. Completed Google Colab/R Notebook containing code, output, comments, and interpretation.**
- 2. PDF version of the completed notebook.**
- 3. SQLite database containing the processed dataset.**
- 4. Three business insights.**
- 5. Upload the complete work to a Git repository and provide the Git repository link in the submission.**

Technical Skills Tested

- CSV, JSON, and Excel data import**
- Data cleaning and preprocessing**
- Missing-value handling**
- dplyr data manipulation**
- left_join() and inner_join()**
- group_by() and summarise()**
- case_when()**
- Business analytics**
- SQLite database connectivity**
- SQL querying**
- Critical reasoning and interpretation**

COs Linked

CO2, CO3 and CO6

LOs Linked

LO 2.2, LO 2.3, LO 3.1, LO 3.2, LO 4.1 and LO 4.2

Workflow

CSV + JSON + Excel → Import & Clean → Integrate → Analyze → SQLite → Business Insights